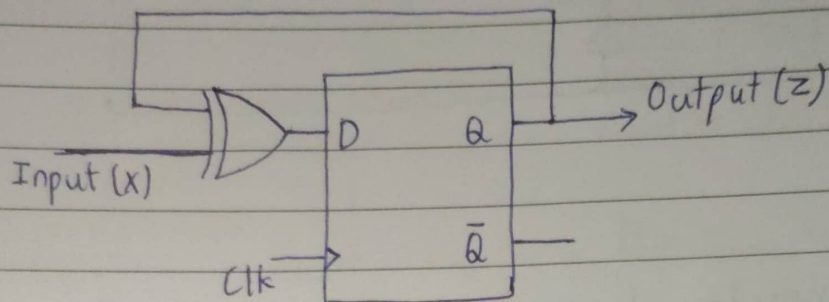


Day-4
Sequential - Assignment

1) Explain the behavior of the given circuit.



$\Rightarrow$ From the above $D = X \oplus Q$ and $Q_{next} = D$

$\therefore$	Q	X	$D = X \oplus Q$	Q_{next}
	0	0	0	0
	0	1	1	1
	1	0	1	1
	1	1	0	0

Comparing input x and the next state Q_{next} we can say that

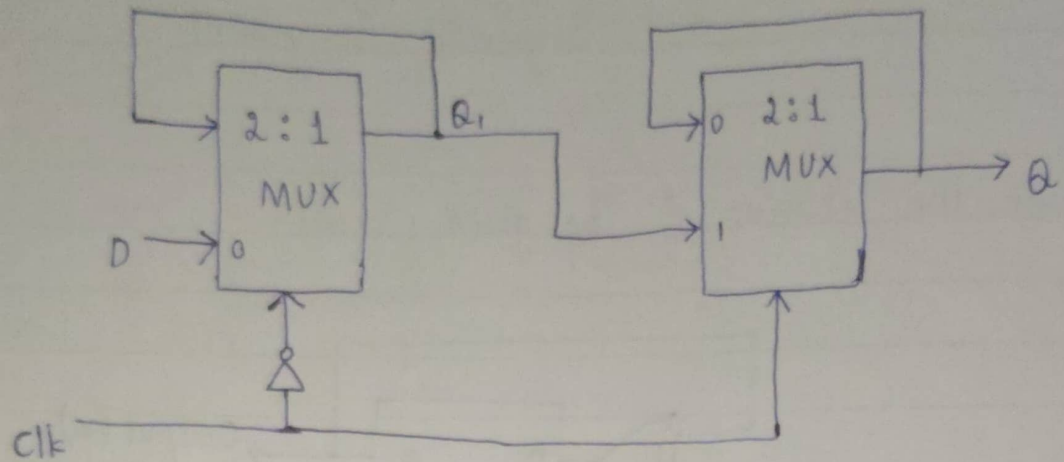
when $x=0$ $Q_{next} = Q$ and

when $x=1$ $Q_{next} = \bar{Q}$

$\therefore$ The behaviour of the given circuit is a T-flipflop.

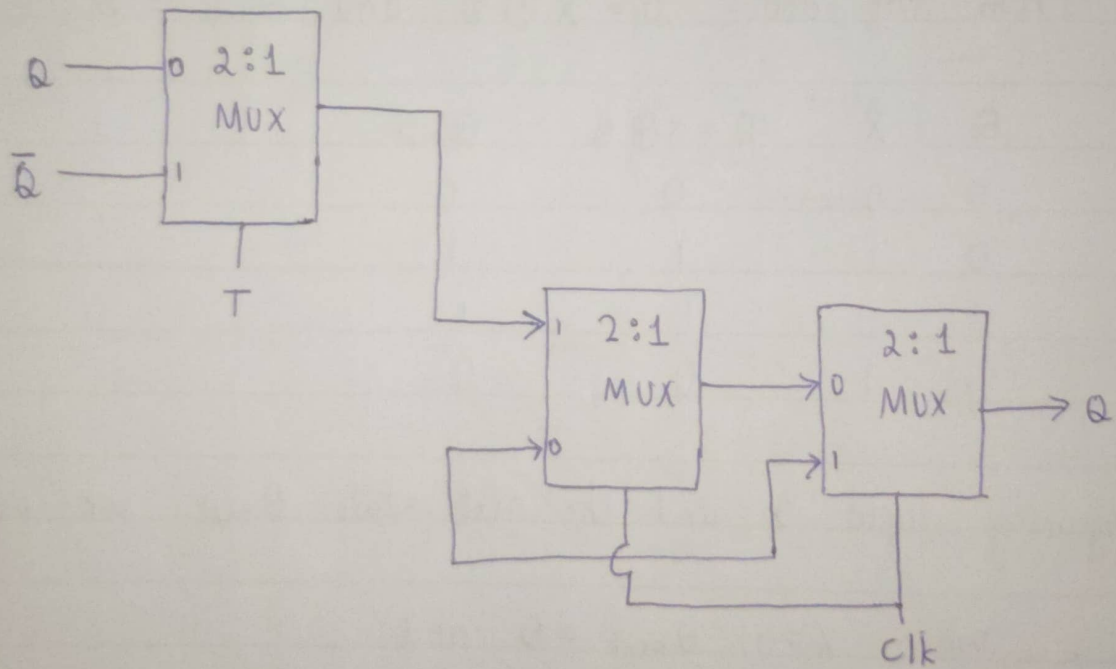
2) Design a posedge triggered D flipflop using 2:1 Muxes only.

=>



3) Design a negedge triggered T flipflop using 2:1 Muxes only.

=>



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u) Convert by adding external gates:

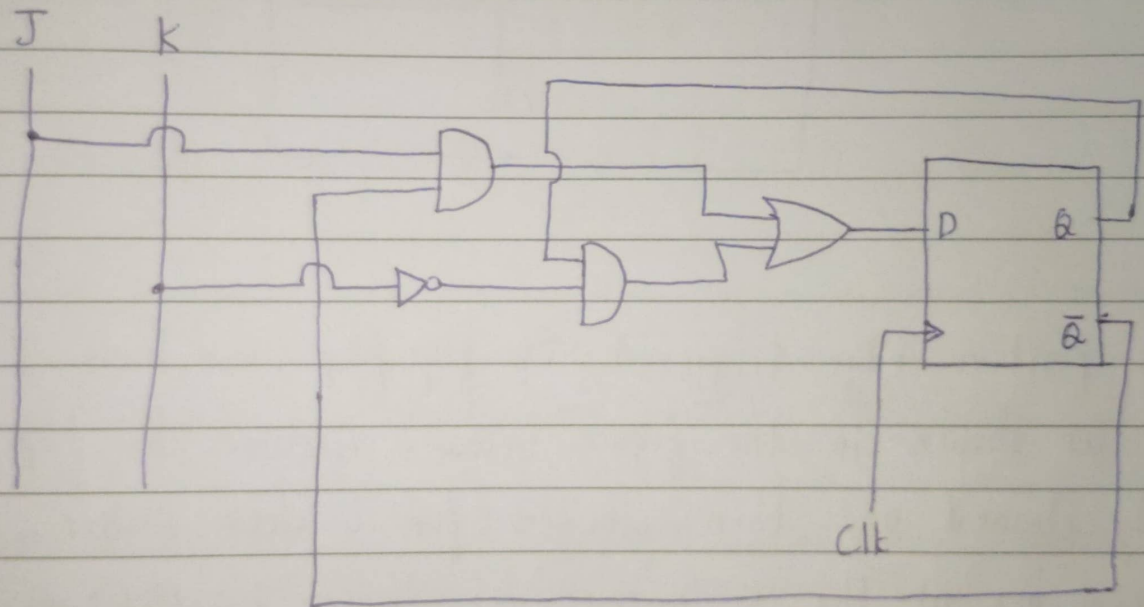
a) A D-flipflop to a J-K flipflop.

⇒

J	K	Present-state (Q)	Q _{next}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

J \ K	0	1	0	1
0	0	1	0	0
1	1	1	0	1

$$D = Q_{next} = \underline{J\bar{Q}} + \underline{\bar{K}Q}$$



b) A T flip-flop to a D flip-flop.

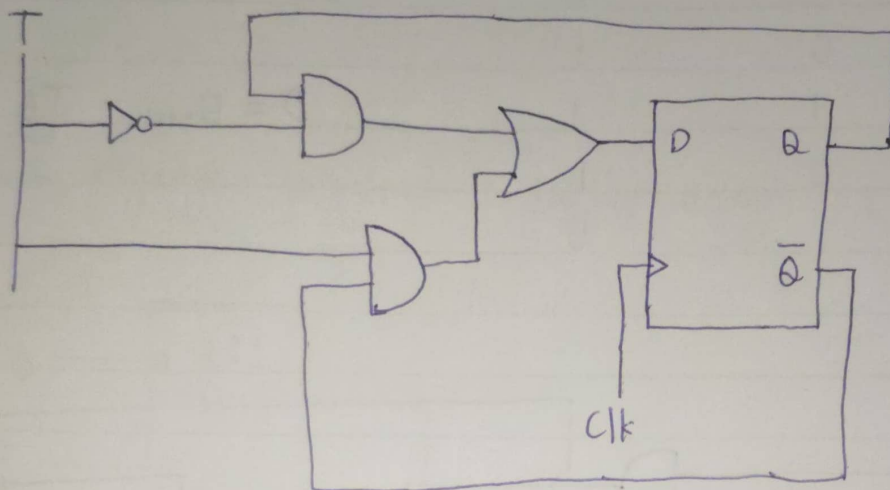
⇒

T	Present Q	Q _{next}
0	0	0
0	1	1
1	0	1
1	1	0

	Q	$\bar{Q}$	Q
T	0	1	1
$\bar{T}$	1	0	0

$$Q_{\text{next}} = \bar{T}Q + T\bar{Q}$$

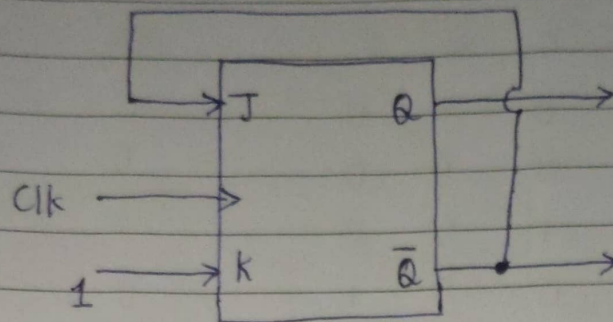
or use XOR gate



5) In a positive edge triggered JK flip flop, we have $J = Q'$ and $K = 1$ as shown in the figure below. Assume the flip flop was initially cleared and then clocked for 6 clock pulses, find the sequence at the Q output for those 6 clock cycles.

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⇒ Positive edge

	J	K	Q	$\bar{Q}$
1 st	1	1	1	0
2 nd	0	1	0	1
3 rd	1	1	1	0
4 th	0	1	0	1
5 th	1	1	1	0
6 th	0	1	0	1

∴ The sequence $Q = \underline{1, 0, 1, 0, 1, 0}$